

a **Method & Evidential Pipeline**

Behavioural experiments

Studies 1a, 1b, and 2



Computational cognitive model

Model the **full utterance distribution** within the Rational Speech Act framework



Held-out prediction

Factorial model comparison
Participant-level PSIS-LOO + posterior prediction

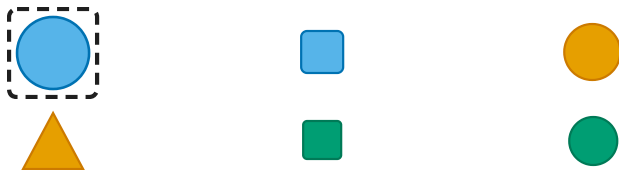
Predictive contribution in observed production

Controlled simulation

Broader range of randomly generated scenes

Generative sufficiency under selected conditions

c **Experimental Design**



“the big blue circle sticker”
target = dashed frame · 6 objects

Factorial design

REFERENTIAL CONTEXT · within participant

Size

size sufficient

Both

both sufficient

Colour

colour sufficient

SIZE DISCRIMINABILITY · between participants

High

strong contrast

Low

weak contrast

Studies 1a/1b

Slider order judgements

Study 2

Direct production: selection, length, and conditional order

b **Two Directions of Communicative Efficiency**

HIERARCHICAL SEMANTIC COMPOSITION

Context-fixed

size meaning

original scene

Size remains anchored to the original scene.

Sequential context updating

size

restricted context

inner adjective + N

An inner property can change the context for interpreting size.

PRAGMATIC PRODUCTION

Global

$\kappa = 0$

big blue N

Complete utterances compete as single plans.

Plan-guided

$0 < \kappa < 1$

advance utterance plan

big

→

blue

→

N

The plan constrains the utterance while successive choices continue to matter.

Fully incremental

$\kappa = 1$

big

→

blue

→

N

Each successive choice is evaluated against prefix-specific alternatives.

All three architectures predict the same distribution over completed responses.

d **Division of Labour**

PRODUCTION ARCHITECTURE

Global

Plan-guided

Fully incremental



$\kappa = 0$

$\kappa = .447$

$\kappa = 1$

best prediction

advance plan

+

successive choices

→

completed response

SEQUENTIAL CONTEXT UPDATING

Sequential context updating

Fitted production
No predictive gain

Controlled simulation
Order asymmetry
with stronger semantic consequences

Across the opposing directions of communicative efficiency, **plan-guided incremental production** best predicts the observed response distribution; **sequential context updating** can generate an order asymmetry under restricted simulation assumptions and adds no predictive gain in the production data.